From two funds to a four-asset portfolio

The idea stays the same. A matrix organizes every pair’s co-movement.

SPY	GLD	MSFT	VCIT
S&P 500 stocks	Gold exposure	One company’s equity	Corporate bonds

1 Daily prices > daily returns

2 Returns > covariance matrix

3 Weights + matrix > portfolio risk

Start with 25% in each asset. What happens to risk when the weights change?

Start with prices on matching dates

Hypothetical SPY and GLD prices • no distributions or external cash flows

Trading day	SPY price	GLD price
1	$100.00	$100.00
2	$102.00	$99.00
3	$100.98	$100.98
4	$104.0094	$100.98

One row represents the same trading day for both assets.

Before calculating: check dates, missing prices and adjustment metadata.

Predict with a partner: on day 2, which asset gained and which lost?

Turn a price change into a daily return

Return expresses the change relative to the money invested yesterday.

Relationship

Simple return = today's price / yesterday's price − 1

Day 2: SPY

102 / 100 − 1 = 0.02 = 2%

Your turn: GLD

Use $100 yesterday and $99 today. Explain the sign.

Repeat for each pair of consecutive prices, for every asset.

The first price has no preceding price: leave its return unavailable.

Build your own formula. Check that a price decline gives a negative return.

Why use LN returns? They add across time

A common modeling convention: convenient for compounding and close to small daily % returns.

One day: $100 to $102 **Simple return = 2.0000%** **LN(102/100) = 1.9803%**

ACROSS TIME • one asset, two days

$100 > $102 > $104.04

Log returns: 1.9803% + 1.9803%

Total log return = 3.9605%

Convert back: EXP(0.039605)-1

Total simple return = 4.04%

ACROSS ASSETS • two assets, one day

50% in A (+2%); 50% in B (−2%)

Portfolio simple return:

50% × 2% + 50% × (−2%) = 0%

Exact portfolio log return = LN(1) = 0

Weighted asset logs are not exact.

Add LOG returns across time. Weight SIMPLE returns across assets.

Returns reveal the pattern that prices hide

Debrief: the same three daily observations will drive every next step.

Trading day	SPY simple return	GLD simple return
2	+2.00%	−1.00%
3	−1.00%	+2.00%
4	+3.00%	0.00%

SPY average daily return = 1.3333%; GLD = 0.3333%.

Volatility measures dispersion around each asset's own average.

Predict: which asset has the greater daily volatility?

Covariance starts with deviations from averages

Subtract each asset's own average • pp = percentage points.

Trading day	SPY deviation	GLD deviation
2	+0.6667 pp	−1.3333 pp
3	−2.3333 pp	+1.6667 pp
4	+1.6667 pp	−0.3333 pp

Above average = positive deviation; below average = negative deviation.

Multiply each same-day pair. Predict the sign of every product.

Discuss with a partner: do these deviations reinforce or offset each other?

Negative covariance reveals offsetting movement

Debrief: all three same-day deviation products are negative.

Pair the deviations

Multiply SPY's deviation by GLD's deviation for each day.

Combine the observations

Sum the products; divide by n − 1, where n = paired days.

Our three-day result

Sample covariance = −0.00026667 (decimal return units squared)

Negative: above-average movements tend to offset below-average movements.

Positive: deviations tend to reinforce. Near zero: little linear co-movement.

Covariance describes joint movement. Its magnitude depends on the assets' volatility.

A matrix organizes every asset pair

Two-asset example • sample estimates from the same daily simple returns

Asset	SPY	GLD
SPY	0.00043333	−0.00026667
GLD	−0.00026667	0.00023333

Diagonal entries = individual asset variances.

Off-diagonal entries = covariance; the two mirrored values match.

For four assets: four variances + six unique covariance pairs.

Weights turn asset returns into portfolio returns

Invest 50% in each asset at the beginning of every day.

Trading day	Weighted calculation	Portfolio return
2	$\frac{1}{2} \times 2\% + \frac{1}{2} \times (-1\%)$	0.50%
3	$\frac{1}{2} \times (-1\%) + \frac{1}{2} \times 2\%$	0.50%
4	$\frac{1}{2} \times 3\% + \frac{1}{2} \times 0\%$	1.50%

Multiply each asset's simple return by its beginning-of-day weight.

Add the contributions. Repeat for each valid trading day.

Predict: will this portfolio fluctuate less than either asset alone?

Portfolio risk is the spread of portfolio returns

Debrief: the combined daily series is 0.50%, 0.50%, 1.50%.

Average daily portfolio return

(0.50% + 0.50% + 1.50%) / 3 = 0.8333%

Measure dispersion around that average

Sample portfolio variance = 0.00003333

Take the square root

Daily portfolio standard deviation = 0.5774%

SPY alone: 2.0817% SD • GLD alone: 1.5275% SD.

The portfolio fluctuates less because the assets' deviations offset.

Risk here means volatility, including upside and downside movement—not maximum loss.

The covariance matrix gives the same portfolio risk

Same daily returns, weights and sample convention > same answer.

Individual-risk contributions

Weight² × SPY variance + weight² × GLD variance

Joint-movement contribution

Add 2 × SPY weight × GLD weight × covariance

Combine, then take the square root

0.00010833 + 0.00005833 − 0.00013333 = 0.00003333

Square root of the daily variance = 0.5774% daily volatility.

For four assets, include all four variances and all six pairwise terms.

Check your model: matrix risk should match risk from weighted SIMPLE daily returns.

Annualize only after calculating daily risk

Keep daily and annual units distinct throughout the model.

Scale variance

Annual variance estimate = daily variance × 252

Scale standard deviation

Annual volatility estimate = daily volatility × √252

Our hypothetical example

0.5774% × √252 = approximately 9.17% per year

This scaling assumes stable daily variance and no serial correlation.

Three observations illustrate the process; use adequate history for estimation.

Annualized volatility is an estimate, not a loss forecast or a guaranteed outcome.

Now construct the analysis independently

Work with a partner; each student builds and checks their own workbook.

Prepare

Identify aligned prices. Choose and explain your return convention.

Estimate

Calculate valid returns, asset variability and pairwise covariance.

Combine and interpret

Choose weights, calculate portfolio risk, then change one allocation.

Keep weights at 100% and use one consistent covariance convention.

Return with your risk estimate, one independent check and an explanation.

Explain why risk changed using weights AND co-movement—not ticker count alone.

Sharpe ratio: excess return per unit of risk

Compare different weight mixes of SPY, GLD, MSFT and VCIT.

Sharpe = (Expected portfolio return – Risk-free rate) / Portfolio standard deviation

Example A: (10% – 4%) / 12% = 0.50 • 0.50 units of excess return per unit of volatility

Hypothetical mix	Annual return	Annual risk (SD)	Sharpe
A	10%	12%	0.50
B	12%	20%	0.40

B earns more return, but A earns more excess return per unit of risk.

Compare using the same dates, annual units and 4% risk-free rate.

Identify your own Sharpe output; use it to compare feasible weight mixes.

Higher Sharpe = better volatility-adjusted return. It does not mean higher return or lower risk.

Use Solver to maximize the Sharpe ratio

Start with equal weights; students identify their own objective and decision cells.

SOLVER PARAMETERS

Objective: maximize

Your portfolio Sharpe ratio

Decision variables

Your four asset weights

Solving method

GRG Nonlinear

SUBJECT TO THE CONSTRAINTS

Total weight = 100%

Fully allocate the portfolio

Each weight ≥ minimum

Use the activity's stated lower bounds

Each weight ≤ maximum

Use the activity's stated upper bounds

Data > Solver > enter settings and Add all three constraints.

Solve > check status and constraints > keep a valid improved solution.

Check all constraints, Solver's status and the change from the starting Sharpe ratio.